1.1.1 ▶ ▶ ▶ ▶ The Kinematics Equations

These five equations, collectively known as the "big five," describe all kinematic motion with constant acceleration. Calculus is often employed to derive the equations, but an algebraic derivation is provided here since these equations are required for AP Physics 1.

To derive the equations, we start with the following definitions:

$$\Delta\phi = \phi_f - \phi_i \text{ for any quantity } \phi$$

$$\phi_{avg} = \frac{\phi_i + \phi_f}{2} \text{ for any quantity } \phi$$

$$v_{avg} = \frac{\Delta x}{\Delta t}$$

$$a_{avg} = \frac{\Delta v}{\Delta t}$$

Consider a particle moving kinematically (with constant acceleration). It's position is represented by x, its velocity v, and its acceleration a. Using the first and last of the four definitions as well as the fact that acceleration is constant[1], we have:

$$\boxed{v_f = v_i + a\Delta t}. \tag{1.1}$$

Using our second definition on our third definition yields:

$$\boxed{\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t}. \tag{1.2}$$

Rearranging eq. 1.2 and plugging it back into 1.1 gives us:

$$\Delta t = \Delta x \left(\frac{2}{v_f + v_i}\right)$$

$$v_f - v_i = \frac{2a\Delta x}{v_f + v_i}$$

$$\boxed{v_f^2 - v_i^2 = 2a\Delta x} \tag{1.3}$$

Plugging 1.1 into 1.2 gives:

$$\Delta x = \frac{1}{2}\Delta t\,(2v_i + a\Delta t)$$

$$\boxed{\Delta x = v_i\Delta t + \tfrac{1}{2}a(\Delta t)^2} \tag{1.4}$$

Lastly, we can substitute v_i from eq. 1.1 into eq. 1.4 to obtain:

$$\Delta x = (v_f - a\Delta t)\Delta t + \tfrac{1}{2}a(\Delta t)^2$$

$$\boxed{\Delta x = v_f\Delta t - \tfrac{1}{2}a(\Delta t)^2} \tag{1.5}$$

[1] i.e. $a_{avg} = a$

Thus, we have the "Big Five" equations, describing all motion with constant acceleration:

$$v_f = v_i + a\Delta t \tag{1.6}$$

$$\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t \tag{1.7}$$

$$v_f^2 - v_i^2 = 2a\Delta x \tag{1.8}$$

$$\Delta x = v_i\Delta t + \tfrac{1}{2}a(\Delta t)^2 \tag{1.9}$$

$$\Delta x = v_f\Delta t - \tfrac{1}{2}a(\Delta t)^2 \tag{1.10}$$

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